

es. 6 foglio III

$$P(E_1) = \frac{1}{3} ; P(E_2) = \frac{1}{4} ; P(E_3) = \frac{5}{12}$$

$A_k = \text{'E}_k \text{ non si presenta in 4 prove'}$
 $k=1, 2, 3$

$$\begin{aligned} P(A_2^c \cap A_3^c) &= P((A_2 \cup A_3)^c) = 1 - P(A_2 \cup A_3) = \\ &= 1 - (P(A_2) + P(A_3) - P(A_2 \cap A_3)) \end{aligned}$$

$$\begin{aligned} P(A_2) &= P(\text{'E}_2 \text{ non si verifica in 4 esperimenti'}) = \\ &= P(E_{2,1}^c \cap E_{2,2}^c \cap E_{2,3}^c \cap E_{2,4}^c) = \prod_{j=1}^4 P(E_{2,j}^c) = \\ &= (1 - p_2)^4 = \left(1 - \frac{1}{4}\right)^4 = \left(\frac{3}{4}\right)^4 \end{aligned}$$

In maniera simile

$$P(A_3) = (1 - p_3)^4 = \left(1 - \frac{5}{12}\right)^4 = \left(\frac{7}{12}\right)^4$$

$$P(A_2 \cap A_3) = P(\text{'si verifica sempre E}_1\text{'}) = p_1^4 = \left(\frac{1}{3}\right)^4$$

$$P(A_2 \cap A_3) = P(\text{'si verificano sempre } E_i\text{'}) = P_i^4 = \left(\frac{1}{3}\right)^4$$

$$\begin{aligned} P(A_2^c \cap A_3^c) &= 1 - (P(A_2) + P(A_3) - P(A_2 \cap A_3)) = \\ &= 1 - \left(\left(\frac{3}{4}\right)^4 + \left(\frac{7}{12}\right)^4 - \left(\frac{1}{3}\right)^4 \right) \end{aligned}$$

Generalizzando ad n prove

$B_k = \text{'non verificarsi di } E_k \text{ su } n \text{ prove'}$

$$\begin{aligned} P(\underline{B_2^c \cap B_3^c}) &= 1 - \left(\left(\frac{3}{4}\right)^n + \left(\frac{7}{12}\right)^n - \left(\frac{1}{3}\right)^n \right) = \\ &= 1 - \left[\frac{3^{2n} + 7^n - 4^n}{12^n} \right] \end{aligned}$$

es. 8

$X = \text{'somma dei 3 numeri estratti'}$

$$X \in \{3, 4, 5, 6, 7, 8, 9\}$$

$$P(X=3) = P('111') = \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{27}$$

$$P(X=9) = P('333') = \frac{1}{27}$$

$$\begin{aligned} P(X=4) &= P('211' \cup '121' \cup '112') = P('211') + P('121') + P('112') \\ &= \frac{1}{27} + \frac{1}{27} + \frac{1}{27} = \frac{3}{27} \end{aligned}$$

$$P(X=8) = P('332' \cup '323' \cup '233') = \frac{3}{27}$$

$$\begin{aligned} P(X=5) &= P('221' \cup '212' \cup '122' \cup '113' \cup '131' \cup '311') \\ &= \frac{6}{27} \end{aligned}$$

$$\begin{aligned} P(X=7) &= P('223' \cup '322' \cup '232' \cup '331' \cup '133' \cup '313') \\ &= \frac{6}{27} \end{aligned}$$

$$\begin{aligned} P(X=6) &= P('222' \cup '123' \cup '213' \cup '231' \cup '321' \cup \\ &\quad \cup '312' \cup '132') = \frac{7}{27} \end{aligned}$$

$$\sum_{k=3}^9 P(X=k) = \frac{1}{27} + \frac{3}{27} + \frac{6}{27} + \frac{7}{27} + \frac{6}{27} + \frac{3}{27} + \frac{1}{27} = \frac{27}{27}$$

$$Y \in \{0, 1, 2\}$$

$$P(Y=0) = \frac{3}{27}$$

$$P(Y=1) = \frac{12}{27}$$

$$P(Y=2) = \frac{12}{27}$$

es. 4

$B_k =$ 'il libro viene trovato nella biblioteca k -esima'

B_k INDEPENDENT $k=1, 2, 3$

$$P(B_1 \cup B_2 \cup B_3) = 1 - P((B_1 \cup B_2 \cup B_3)^c) =$$

$$= 1 - P(B_1^c \cap B_2^c \cap B_3^c) =$$

$$= 1 - P(B_1^c)P(B_2^c)P(B_3^c) = 1 - \left(\frac{3}{4}\right)^3$$

$$= 1 - P(B_1^c) P(B_2^c) P(B_3^c) = 1 - \left(\frac{3}{4}\right)^3$$

$$P(B_1^c) = P(B_2^c) = P(B_3^c) = 1 - P(B_i) = 1 - \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$$

VALOR MEDIO o VALORE ATTESO o MEDIA TEORICA o SPERANZA MATEMATICA ...

Def. Per una v.c. X discreta
con $X \in \{x_1, \dots, x_n\}$, il valor medio, se esiste,

$$E[X] := \sum_{k=1}^n x_k p(x_k).$$

Per una v.c. X continua il valor medio,
se esiste, è dato da

$$E[X] := \int_{-\infty}^{+\infty} x f(x) dx.$$

es. 1 v.c. di Bernoulli:

$$X = \begin{cases} 1 & \text{se si verifica } A \text{ in un} \\ & \text{esperimento} \\ 0 & \text{altrimenti} \end{cases}$$

$$P(X=1) = P(A) = p$$

$$P(X=0) = P(A^c) = 1 - p = q$$

$$X \sim \text{Be}(p)$$

$$E[X] := \sum_{k=1}^n x_k p(x_k) = 0 \cdot q + 1 \cdot p = p$$

es. 2 X = 'esito lancio di un dado cubico
non truccato'

$$X \in \{1, 2, 3, 4, 5, 6\}$$

$$p(k) = P(X=k) = \frac{1}{6} \quad \text{se } k = 1, 2, \dots, 6$$

$$E[X] := \sum_{k=1}^6 x_k p(x_k) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} =$$

$$+ 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} =$$

$$= \frac{1}{6} (21) = \frac{7}{2} = 3.5$$

es. 3 X v.c. continua con

$$f(x) = \begin{cases} \frac{1}{2} & \text{se } x \in [0, 2] \\ 0 & \text{altrove} \end{cases}$$

$$E[X] = \int_{-\infty}^{+\infty} x f(x) dx = \int_{-\infty}^0 x \cdot 0 dx + \int_0^2 x \cdot \frac{1}{2} dx + \int_2^{+\infty} x \cdot 0 dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} \right]_0^2 = \frac{1}{2} \cdot \frac{4}{2} = 1$$

PROPRIETÀ

1) Se X è una v.c. nota e $Y = g(X)$, allora

$$E[Y] = E[g(X)] = \begin{cases} \sum_{k=1}^n g(x_k) p(x_k) & \begin{array}{l} X \text{ v.c. discreta} \\ X \in \{x_1, \dots, x_n\} \end{array} \\ \int_{-\infty}^{+\infty} g(x) f(x) dx & \rightarrow \dots \text{ continua} \end{cases}$$

$$\int_{-\infty}^{+\infty} g(x) f(x) dx$$

X v.c. continue
con $f(x)$
funzione di
densità

(se esiste $E[Y]$)

2) Sia X v.c. nota e $Y = \alpha X + \beta$ con $\alpha, \beta \in \mathbb{R}$
se esistono $E[X]$ ed $E[Y]$

$$E[Y] = \alpha E[X] + \beta$$

DIM X v.c. discreta con $X \in \{x_1, \dots, x_n\}$

$$Y = g(X)$$

$$E[Y] = \sum_{k=1}^n g(x_k) p(x_k) = \sum_{k=1}^n (\alpha x_k + \beta) p(x_k) =$$

$$= \sum_{k=1}^n (\alpha x_k p(x_k) + \beta p(x_k)) =$$

$$= \sum_{k=1}^n \alpha x_k p(x_k) + \sum_{k=1}^n \beta p(x_k) =$$

$$= \sum_{k=1}^n \alpha x_k p(x_k) + \sum_{k=1}^n \beta p(x_k) =$$

$$= \alpha \sum_{k=1}^n x_k p(x_k) + \beta \sum_{k=1}^n p(x_k) =$$

$\parallel E[X] \text{ per def.}$
 $\parallel 1$

$$= \alpha E[X] + \beta \quad \square$$

in modo simile
si dimostra il caso
continuo.

3) Dato una coppia di v.c. (X, Y) e definita

$Z = h(X, Y)$, allora, se esiste,

$$E[Z] = E[h(X, Y)] = \begin{cases} \sum_{k=1}^n \sum_{j=1}^m h(x_k, y_j) p(x_k, y_j) \\ \text{se } (X, Y) \text{ v.c. discrete con} \\ X \in \{x_1, \dots, x_n\}, Y \in \{y_1, \dots, y_m\} \\ \\ \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x, y) f(x, y) dx dy \end{cases}$$

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x,y) f(x,y) dx dy = \sigma$$

(X,Y) v.c. congiuntamente continue

4) Dato una coppia di v.c. (X,Y) , allora, se esiste,
 $E[X+Y] = E[X] + E[Y]$

Dim) (X,Y) v.c. congiuntamente continue

$$E[X+Y] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x+y) f(x,y) dx dy =$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x,y) dx dy + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x,y) dx dy =$$

$$= \int_{-\infty}^{+\infty} x \left(\int_{-\infty}^{+\infty} f(x,y) dy \right) dx + \int_{-\infty}^{+\infty} y \left(\int_{-\infty}^{+\infty} f(x,y) dx \right) dy =$$

$$= \int_{-\infty}^{+\infty} x f_X(x) dx + \int_{-\infty}^{+\infty} y f_Y(y) dy =$$

$$= \int_{-\infty}^{\infty} x f_X(x) dx + \int_{-\infty}^{\infty} y f_Y(y) dy$$

in maniera simile
si dimostra il
caso discreto

N.B. la proprietà 4 può essere estesa
alla somma di n v.c.

$$E\left[\sum_{k=1}^n X_k\right] = \sum_{k=1}^n E[X_k]$$

es. N fatture e N indirizzi email.
Sceita una fattura, essa viene inviata
ad un indirizzo scelto a caso.

Quale è il numero medio di fatture
che arrivano all'indirizzo corretto?

$$X_k \sim \text{Be}(p)$$

$$X_k = \begin{cases} 1 & \text{se la } k\text{-esima fetta} \\ & \text{è rivolta all'indirizzo} \\ & \text{corretto} \\ 0 & \text{altrimenti} \end{cases}$$

$$P(X_k=1)=p=\frac{1}{N}$$

$$E[X_k]=p=\frac{1}{N}$$

Y = 'n° fetture che arrivano all'indirizzo corretto'

$$Y = \sum_{k=1}^N X_k$$

$$E[Y] = E\left[\sum_{k=1}^N X_k\right] = \sum_{k=1}^N E[X_k] = \sum_{k=1}^N \frac{1}{N} = N \cdot \frac{1}{N} = 1$$

VARIANZA DI UNA V.C.

Def. Dato una v.c. X con valor medio $\mu = E[X]$

... e' 212 una v.c. con valore medio $\mu = \dots$
si dice varianza di X , se esiste,

$$\text{Var}(X) := E[(X - \mu)^2]$$

OSS. $\text{Var}(X) = \sum_{k=1}^n \underbrace{(x_k - \mu)^2}_{\text{green}} p(x_k)$ nel caso di X
v.c. discreta
 $X \in \{x_1, \dots, x_n\}$

$$\text{Var}(X) = \int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx$$
 nel caso di X
v.c. continua

PROPRIETÀ

1) $\text{Var}(X) \geq 0 \quad \forall X \text{ v.c.}$

$$2) \text{Var}(\alpha X + \beta) = \alpha^2 \text{Var}(X) \quad \text{se } \alpha, \beta \in \mathbb{R}$$

$$\boxed{\text{Dim}} \quad Y = \alpha X + \beta, \quad E[Y] = \alpha E[X] + \beta \underset{\substack{\uparrow \\ \mu = E[X]}}{=} \alpha \mu + \beta$$

$$\begin{aligned} \text{Var}(\alpha X + \beta) &= \text{Var}(Y) := E[(Y - E[Y])^2] = \\ &= E[(\cancel{\alpha X + \beta} - (\cancel{\alpha \mu + \beta}))^2] = \\ &= E[(\alpha X - \alpha \mu)^2] = E[\alpha^2 (X - \mu)^2] = \\ &= \alpha^2 E[(X - \mu)^2] = \alpha^2 \text{Var}(X) \quad \square \end{aligned}$$

N.B. Se $\alpha = 0$ e $\beta \neq 0$

$$\text{Var}(\beta) = 0$$

questo risultato è in
accordo con il significato
matematico della varianza

$$3) \text{Var}(X) = E[X^2] - E[X]^2$$

$$\boxed{\text{Dim}} \quad E[X] = \mu$$

$$\text{Var}(X) := E[(X - \mu)^2] = E[X^2 + \mu^2 - 2\mu X] =$$

$$= E[X^2] + E[\mu^2] + E[-2\mu X] =$$

$\swarrow$
 proprietà
 di $E[\cdot]$

$$= E[X^2] + \mu^2 - 2\mu E[X] =$$

$$= E[X^2] + \mu^2 - 2\mu^2 = E[X^2] - \mu^2 =$$

$$= E[X^2] - (E[X])^2 \quad \square$$

Def MOMENTO DI ORDINE n DI X

Dato una variabile casuale X si dice momento
 $1. \quad 1. \quad \dots \quad 1. \quad \vee \quad \dots \quad 1.$

Dato una variabile casuale X si dice momento di ordine n di X , se esiste,

$$E[X^n] \quad n \in \mathbb{N} \setminus \{0\}$$

4) $\text{Var}(X+Y) = ?$

$$Z = X + Y \quad E[Z] = E[X] + E[Y]$$

$$\mu_X = E[X]; \mu_Y = E[Y] \Rightarrow E[Z] = \mu_X + \mu_Y$$

$$\begin{aligned} \text{Var}(Z) &:= E[(Z - E[Z])^2] = E[(X + Y - (\mu_X + \mu_Y))^2] = \\ &= E[\{(X - \mu_X) + (Y - \mu_Y)\}^2] = \\ &= E[(X - \mu_X)^2 + (Y - \mu_Y)^2 + 2(X - \mu_X)(Y - \mu_Y)] = \end{aligned}$$

$$= \underbrace{E[(X - \mu_X)^2]} + \underbrace{E[(Y - \mu_Y)^2]} + E[2(X - \mu_X)(Y - \mu_Y)] =$$

$$= \text{Var}(X) + \text{Var}(Y) + 2 \underbrace{E[(X - \mu_X)(Y - \mu_Y)]}_{\text{Cov}(X, Y)} =$$

$$= \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

Covarianza di X e Y

⇓

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

es. 1 $X \sim \text{Be}(p)$ $E[X] = p$

$\text{Var}(X)$? $\text{Var}(X) = E[X^2] - E[X]^2$

$$E[X^2] = \sum_{k=0}^2 x_k^2 p(x_k) = 0^2 p(0) + 1^2 p(1) = p(1) = p$$

$$E[X^2] = \sum_{k=1}^{\infty} x_k^2 p(x_k) = 0^2 p(0) + 1^2 p(1) = p(1) = p$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = p - p^2 = p(1-p) = p \cdot q$$

es. 2 lancio d'2do cubico non truccato

$$X \in \{1, \dots, 6\} \quad E[X] = \frac{7}{2}$$

$$\text{Var}(X) = E[X^2] - E[X]^2$$

$$E[X^2] = \sum_{k=1}^6 x_k^2 p(x_k) = \sum_{k=1}^6 k^2 \cdot \frac{1}{6} = \frac{1}{6} (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) =$$

$$= \frac{1}{6} (1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6}$$

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182 - 147}{12} = \frac{35}{12}$$

es. 3 X v.c. continua $f(x) = \frac{1}{2}$ se $x \in [0, 2]$

es. 3 X v.c. continuu $f(x) = \begin{cases} \frac{1}{2} & \text{se } x \in [0, 2] \\ 0 & \text{altrove} \end{cases}$

$$E[X] = 1$$

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_{-\infty}^0 x^2 \cdot 0 dx + \int_0^2 x^2 \cdot \frac{1}{2} dx + \int_2^{+\infty} x^2 \cdot 0 dx =$$

$$= \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{4}{3} - 1 = \frac{1}{3}$$

Oss $\text{Cov}(X, Y) := E[(X - \mu_X)(Y - \mu_Y)]$

$$\Rightarrow \text{Cov}(X, X) = \text{Var}(X) = E[(X - \mu_X)(X - \mu_X)]$$